

23 Initially:

$$I = \frac{V_{\text{LDR}}}{r_{\text{LDR}}} = \frac{4.5}{5.3}$$

$$V_R = 10 - 4.5 = 5.5 \text{ V}$$

$$R = \frac{V_R}{I} = \frac{5.5}{4.5 \div 5.3} = 6.48 \, \Omega$$

by potential divider rule:

$$\frac{V_{\text{LDR}}}{\text{e.m.f.}} = \frac{r_{\text{LDR}}}{r_{\text{LDR}} + R}$$

$$\begin{aligned} V_{\text{LDR}} &= (\text{e.m.f.}) \frac{r_{\text{LDR}}}{r_{\text{LDR}} + R} \\ &= (10) \left(\frac{3.1}{3.1 + 6.48} \right) \\ &= 3.2 \text{ V} \end{aligned}$$

24 number of turns